

239 - Sliding Window Maximum

```
Input: nums = [1,3,-1,-3,5,3,6,7], k = 3
```

```
Output: [3,3,5,5,6,7]
```

Explanation:

Window position

$$\begin{bmatrix} 1 & 3 & -1 \end{bmatrix} \begin{bmatrix} -3 & 5 & 3 & 6 & 7 \end{bmatrix}$$

```
1 [3 -1 -3] 5 3 6 7
```

```
1 3 [-1 -3 5] 3 6 7
1 3 [-1 -3 5] 3 6 7
```

$$\begin{array}{ccccccc} 1 & 3 & -1 & [-3 & 5 & 3] & 6 & 7 \\ 1 & 3 & 1 & 3 & [5 & 3 & 4] & 7 \end{array}$$
$$\begin{array}{ccccccc} 1 & 3 & -1 & -3 & [5 & 3 & 6] & 7 \\ 1 & 3 & -1 & -3 & 5 & [3 & 6 & 7] \end{array}$$

$$|res| = |nums) - k + 1$$

13-1

3 -

queve

3 3 -1

33-1

131205

$$3, 1 \quad (1, 3, 1)$$
 $3,2 \quad (3,1,2)$
$$2, 0 \quad (1, 2, 0)$$

queue - monotonically not increasing stack

3, 3, 1

3, 2

⑥ Adding element to the queue

۱, ۳, ۵, ۷

Handwritten numbers 3 and 1. The number 3 is formed by two strokes: a top curve and a bottom curve. The number 1 is formed by two strokes: a vertical line and a horizontal base.

```
while q and q[-1] < nums[i]
```

q.popLeft()

① Removing expiring elems not in window

LIFO

`q[0] == nums[i - k]:`

q. popLeft()